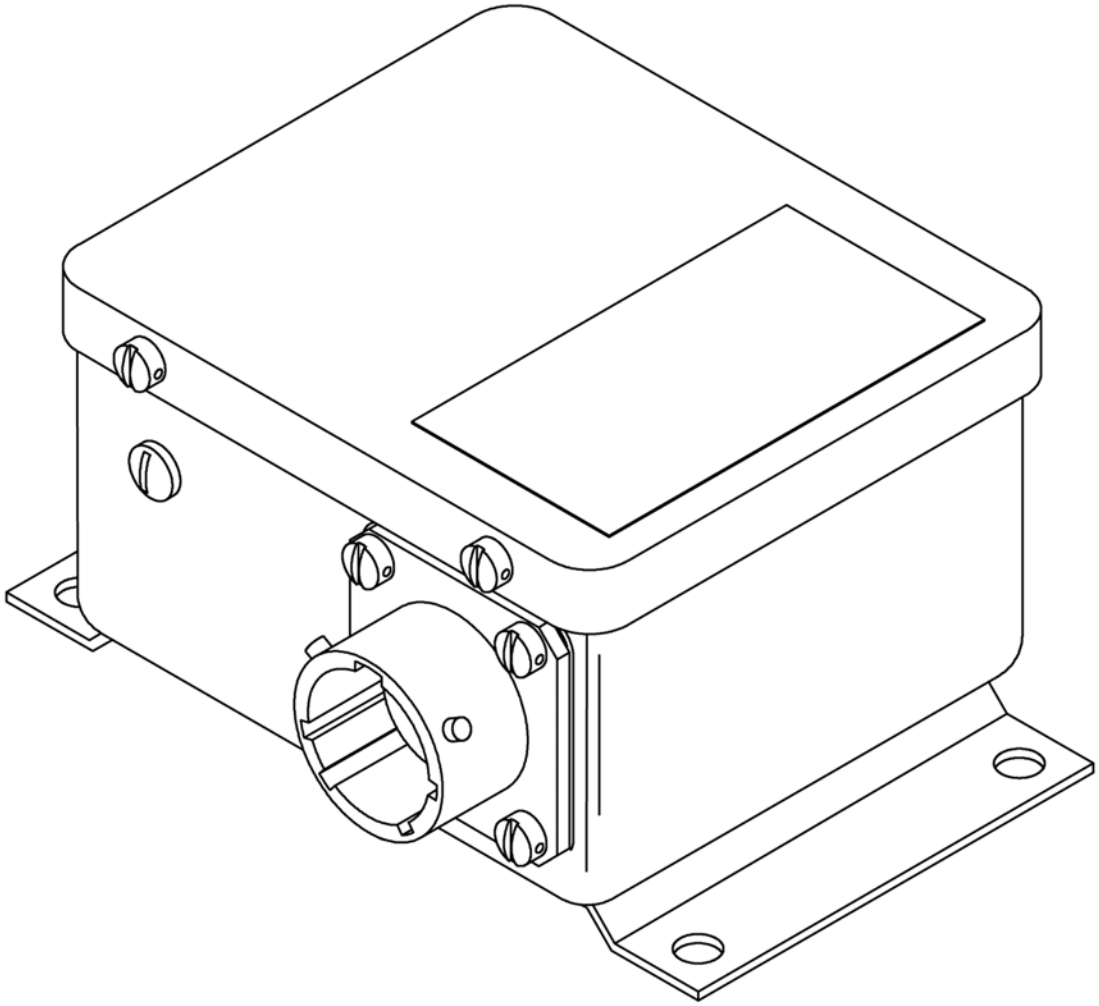
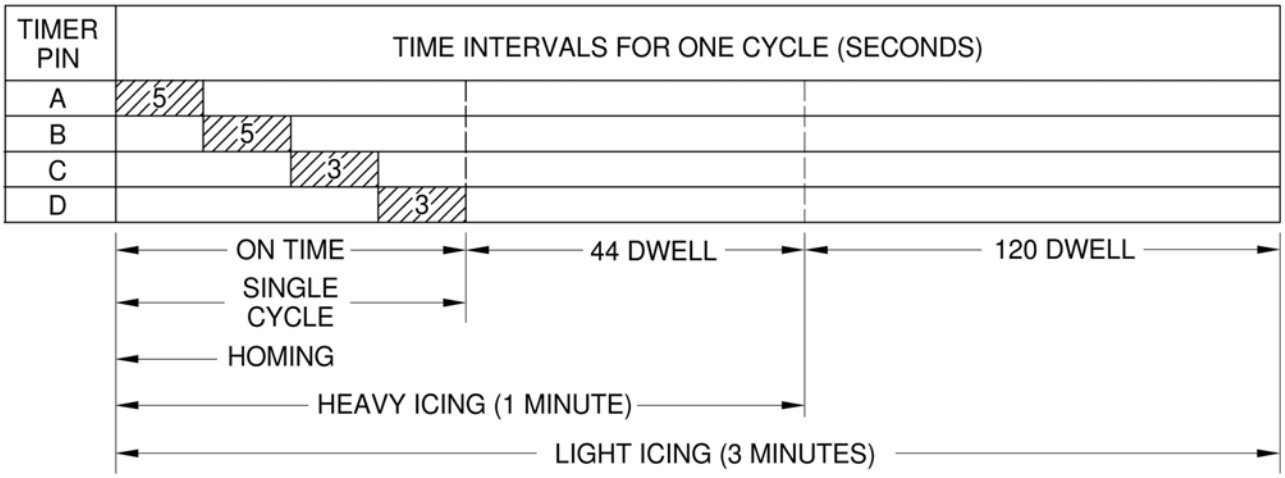
Electronic Timer (Mod S.O.O. 6004) – General Data

1. General

The Goodrich pneumatic timer provides the energizing voltage to the wing and tail de-icing distributor valves, in the system introduced by S.O.O. 6004. The timer, located in the left side of the cabin roof between fuselage station frames 200.36 and 219.32, provides a five second “on” period for the two solenoids in each wing distributor valve, and a three second “on” period for both solenoids in the tail distributor valve. Cycling can be controlled at a fast or slow rate depending on the selection of BOOTS AUTO – FAST – SLOW switch. The total cycling time (including dwell time) is 60 seconds when selected to FAST, and 180 seconds at SLOW. The timer (Figure 1), consists of a metal box with an external connector receptacle, which houses a circuit selector, component board and the necessary electrical wiring. The box has a cover which is sealed with silicon rubber and secured with four screws, it also has two flanges which are drilled to facilitate mounting the complete assembly.



Pneumatic De-icing Timer
Figure 1



NOTES:

1. SHADED AREAS REPRESENT CONTROL INTERVAL TIMES.
2. CYCLING IS CONTINUOUS AND STARTS AT THE BEGINNING OF THE CONTROL INTERVALS.
3. TIME INTERVALS DO NOT OVERLAP.

Timing Chart
Figure 2

2. Servicing

Since the pneumatic de-icing timer is an electronic assembly the servicing excludes any repair according to the manufacturer. Any repair and replacement of component parts must be done only through authorized repair and overhaul facilities.

3. Adjustment/Test

A. Test Equipment

In addition to the required test equipment listed in Table 1 — Test Equipment, a bench test circuit must be available to conduct an operational check of the timer. Individual components of the bench test circuit are listed in Table 3 — Bench Test Circuit Components. The bench test circuit is illustrated in Figure 3.

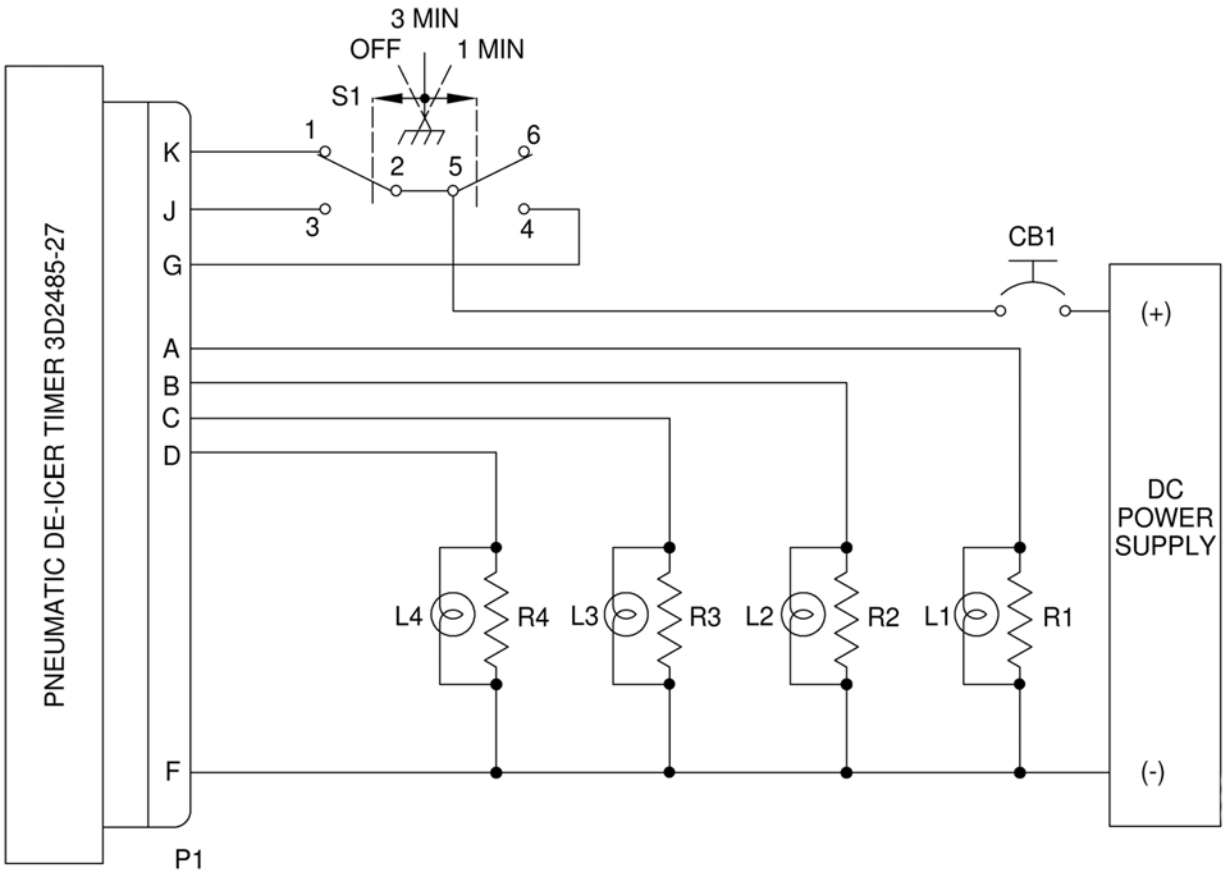
Note Equivalent substitutes may be used.

Table 1 Test Equipment

Item	Identification	Source (CAGE Code)
Bench test circuit		Local fabrication. Refer to Table 3 — Bench Test Circuit Components for individual components
DC power supply; variable, regulated	0 to 32 Volt DC, 5 amps	Local purchase
Insulation resistance tester	Model 225 (500 Volt DC)	Slaughter Co.; 8011 Hailey St.; Ardmore, OK 73401 (V05611)

Table 1 Test Equipment (continued)

Item	Identification	Source (CAGE Code)
Mating electrical connector (with terminating test leads)	MS3116E12–10S	Local purchase
Stopwatch		Local purchase



Timer Test Circuit
Figure 3

4. Inspection/Check

A. Procedure

1) Visually examine the timer for:

a) Signs of corrosion.

b) Loose or missing sealant.

c) Nicks, cracks and dents.

d) Distortion.

e) Legible label.

2) Shake the timer and check for the sound of loose parts inside the case.

3) Examine black sealant for a continuous fillet around the case-to-lid seam and around the connector. If required, reapply sealant to damaged areas.

4) Visually examine electrical connector for:

a) Loose, broken, or shorted lead wire connectors.

b) Electrical arcing paths.

c) Loose or defective attaching parts.

d) Corroded contacts and terminals.

e) Bent and damaged pins.

f) Damage caused by excessive heat.

B. Insulation Resistance Test

WARNING **THE INSULATION RESISTANCE TESTER CAN OUTPUT POTENTIALLY LETHAL VOLTAGES. AVOID CONTACT WITH THE CONNECTION POINT ON TESTER AND PROBE LEAD WIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRICAL SHOCK OR DEATH.**

Do the insulation resistance check before doing the operational tests so that any circuit damage due to this check can be detected during the functional testing that follows.

1) Using a mating electrical connector with terminating test leads, connect test leads from all electrical connector pins together, forming a wire bundle.

2) Ensure that insulation resistance tester is OFF.

3) Connect one lead of insulation resistance tester to wire bundle.

4) Connect other lead of insulation resistance tester to external screw on timer.

5) Apply power to insulation resistance tester.

6) Apply 500 (±25) Volt DC between wire bundle and external screw on timer.

7) Observe that resistance is greater than 100 megohms.

8) Turn insulation resistance tester OFF.

9) Disconnect insulation resistance tester from wire bundle and external screw on timer.

10) Remove mating electrical connector from timer.

C. Operational Tests

1) Operational Test Setup

a) With circuit breaker CB1 and switch S1 in OFF position, connect timer to bench test circuit as shown in Figure 3.

b) Apply power to power supply, and adjust power supply output to 18 (±1) Volt DC.

c) Close circuit breaker CB1.

2) Time Output Checks

a) Single-Cycle Check

1 Momentarily set switch S1 to either the 1 MIN or 3 MIN position and then back to the OFF position, and time output intervals with stopwatch.

2 Observe that lamp operate as shown in the following chart.

Lamp	Time (seconds)			
L1	5			
L2		5		
L3			3	
L4				3

3 After 16 seconds, observe that timer rapidly “homes” in a series of audible clicks, and then stops.

b) One Minute Mode Check

1 Set switch S1 to 1 MIN position, and time output intervals with stopwatch.

2 Observe that lamps operate as shown in the following chart.

Lamp	Time (seconds)				
L1	5				
L2		5			
L3			3		
L4				3	
	60				

c) Three Minute Mode Check

1 Set switch S1 to 3 MIN position, and time output intervals with stopwatch.

2 Observe that lamps operate as shown in the following chart.

Lamp	Time (seconds)				
L1	5				
L2		5			
L3			3		
L4				3	
	180				

D. Special Tools Fixtures and Equipment

The following items are required for testing of the timer. Standard shop tools and consumables required for these functions are not listed.

In addition to the test equipment listed in Table 2 — Special Tools, Fixtures, and Equipment, a bench test circuit must be available to conduct an operational check of the timer. Individual components of the bench test circuit are listed in Table 3 — Bench Test Circuit Components. The bench test circuit is illustrated in Figure 3.

Note Equivalent subtitles may be used for listed items.

Table 2 Special Tools, Fixtures, and Equipment

Item	Identification	Source (CAGE Code)
Bench test circuit		Local Fabrication. Refer to Table 3 — Bench Test Circuit Components for individual components
DC power supply; variable, regulated	0 to 32 Volt DC, 5 amps	Local purchase
Insulation resistance tester	Model 225 (500 Volt DC)	Slaughter Co.; 801 Hailey St.; Ardmore, OK 73401 (V05611)
Mating electrical connector (with terminating test leads)	MS3116E12–10S	Local purchase
Stopwatch		Local purchase

Table 3 Bench Test Circuit Components

Reference Designator (Refer to Figure 3)	Component	Description/Identification	Source (CAGE Code)
CB1	Circuit breaker	5 amp	Local purchase
L1 through L4	Lamp	28 Volt DC , MS25041–6–327	Local purchase
P1	Plug	MS3116E12–10S	Local purchase
R1 through R4	Resistor	50 watt, 20 ohm	Local purchase
S1	Switch, toggle	Double-pole, Double-throw, MS27407–4	Local purchase
–	Wiring	20 gauge	Local purchase